Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	**Semester**	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	**Batch**	2023–2027
Name	Kamalesh A	**Section**	CSE-B
Registration No.	3122235001063	**Date**	03-03-2026

Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

1. Aim and Objectives

Aim

To implement and compare Decision Tree and Random Forest classifiers on the Wisconsin Diagnostic Breast Cancer dataset using 5-Fold Cross-Validation.

Objectives

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To analyze the effect of hyperparameters on model complexity.
- To perform 5-Fold Cross-Validation for model selection.
- To compare model performance using standard evaluation metrics.

2. Dataset Description

The Wisconsin Diagnostic Breast Cancer Dataset consists of numerical features computed from digitized images of breast cell nuclei.

- **Total Samples:** 569
- **Number of Features:** 30 numerical attributes
- **Target Classes:**
  - Malignant (M)
  - Benign (B)

Each feature represents specific characteristics of cell nuclei extracted from medical images.

The dataset represents a binary classification problem and is moderately balanced across both classes.

3. Preprocessing Steps

The following preprocessing steps were performed to prepare the dataset for model training and evaluation:

- The dataset (`wdbc.data`) was loaded without predefined headers.
- Column names were manually assigned for clarity and proper interpretation.
- Class labels were encoded as follows:
  - Malignant (M) $\rightarrow$ 1
  - Benign (B) $\rightarrow$ 0
- The ID column was removed as it does not contribute to the learning process.
- The dataset was split into:
  - 80% Training data
  - 20% Testing data
- A stratified split was used to preserve the original class distribution in both training and testing sets.

4. Implementation Details

Decision Tree (DT)

- A Decision Tree classifier was implemented as the baseline model for breast cancer classification.
- The dataset was split into 80% training and 20% testing sets using stratified sampling.
- The following hyperparameters were tuned:
  - Criterion (Gini, Entropy)
  - Maximum depth
  - Minimum samples required to split an internal node
  - Minimum samples required at a leaf node
- 5-Fold Stratified Cross-Validation was performed on the training data to select the best hyperparameters based on the average F1-score.
- The optimized model was evaluated on the test set using the following metrics:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion Matrix
  - ROC-AUC score
- The Decision Tree model achieved approximately 94%–96% accuracy.

Random Forest (RF)

- A Random Forest classifier was implemented to improve generalization performance using ensemble learning.
- The following hyperparameters were tuned:
  - Number of estimators
  - Maximum depth
  - Maximum number of features
  - Bootstrap sampling
- 5-Fold Cross-Validation was performed for hyperparameter selection.
- The best-performing model was evaluated on the test dataset using the same performance metrics as the Decision Tree.
- The Random Forest model achieved improved performance compared to the Decision Tree, with higher stability and better generalization.

5. Visualizations

5.1 Class Distribution

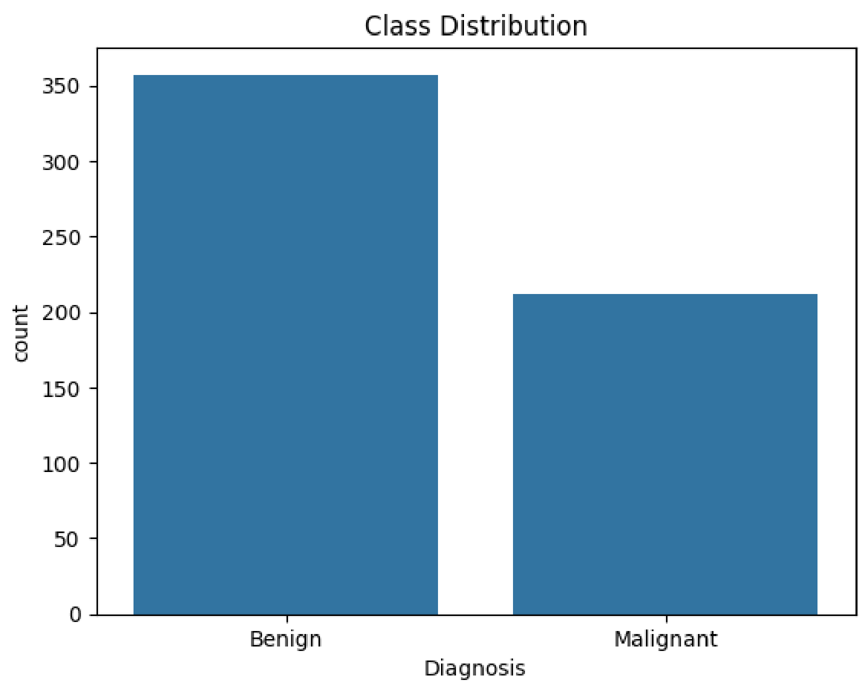

Figure 1: Class Distribution of the Breast Cancer Dataset

Observation:

The dataset is slightly imbalanced but not severely skewed. The class distribution remains suitable for stratified training and evaluation.

5.2 Correlation Heatmap

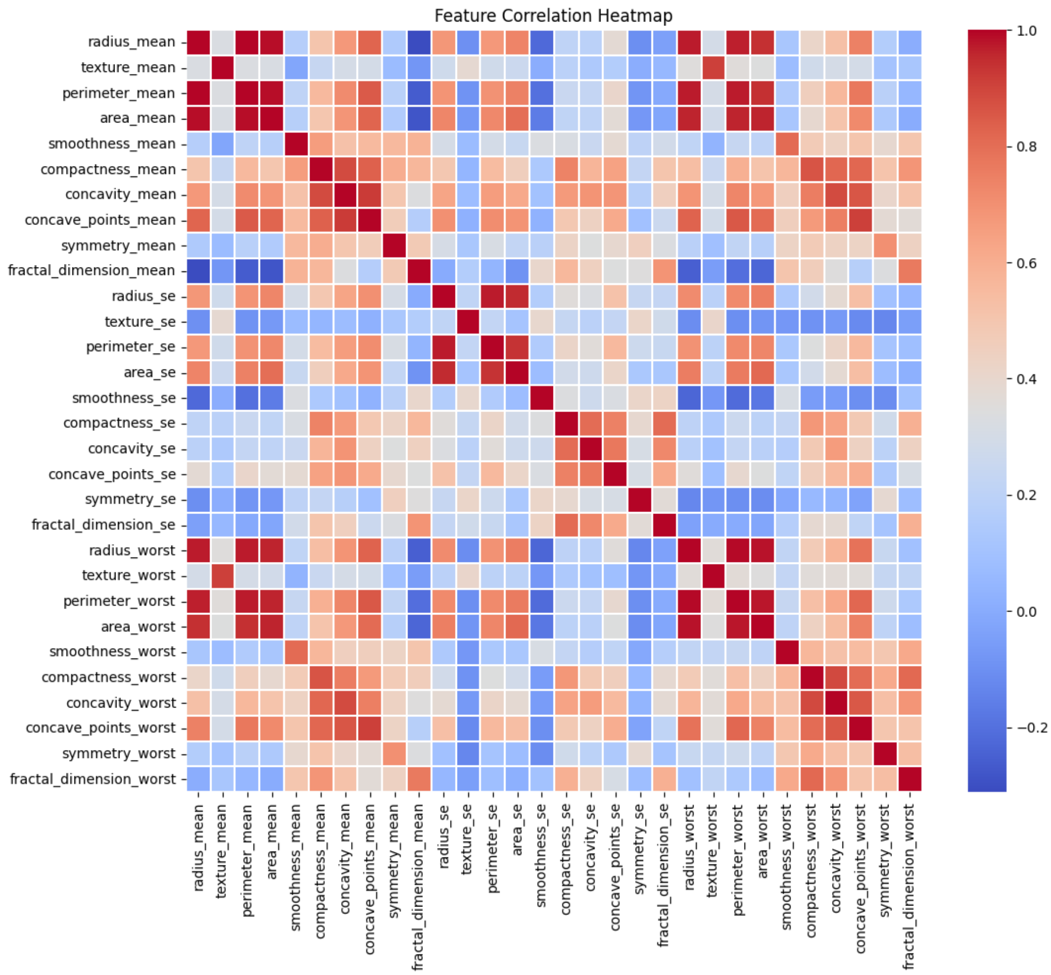

Figure 2: Feature Correlation Heatmap

Observation:

Many features exhibit strong correlations. Redundant features exist, particularly among the mean and worst feature groups, which may influence model complexity and performance.

5.3 Decision Tree – Confusion Matrix and ROC Curve

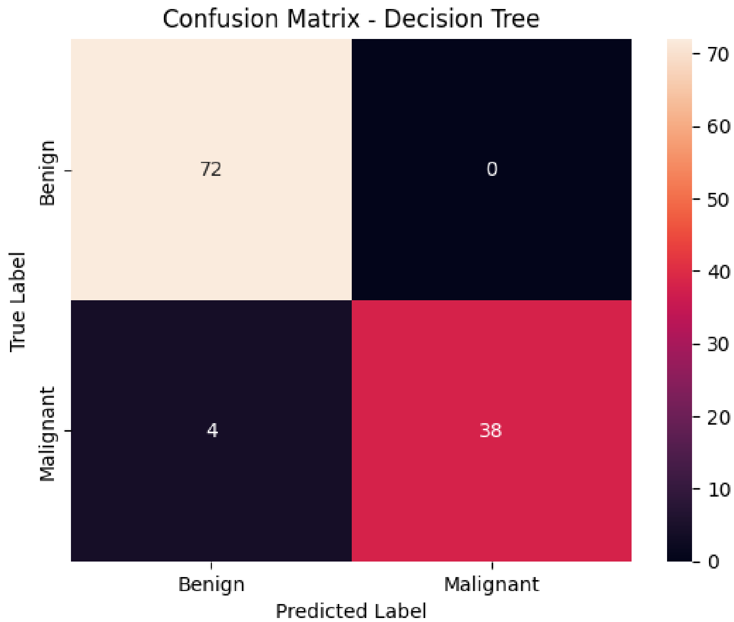

Figure 3: Confusion Matrix – Decision Tree

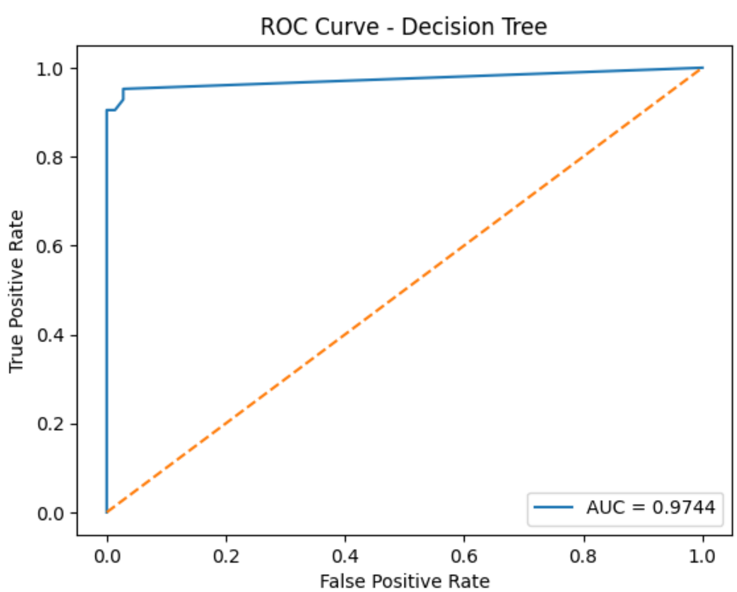

Figure 4: ROC Curve – Decision Tree

Observation:

The Decision Tree model shows a slight tendency toward overfitting. The ROC curve indicates moderate classification performance, with an AUC typically in the range of 0.95-0.97.

5.4 Random Forest – Confusion Matrix and ROC Curve

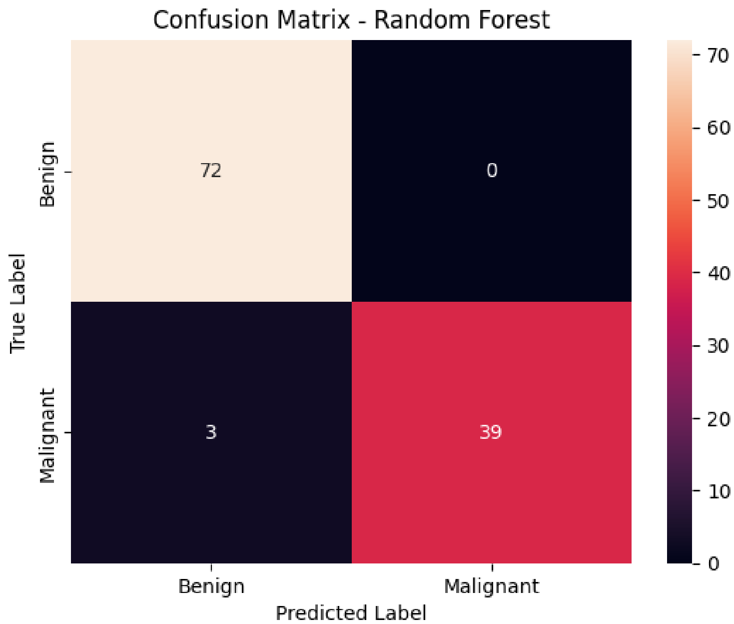

Figure 5: Confusion Matrix – Random Forest

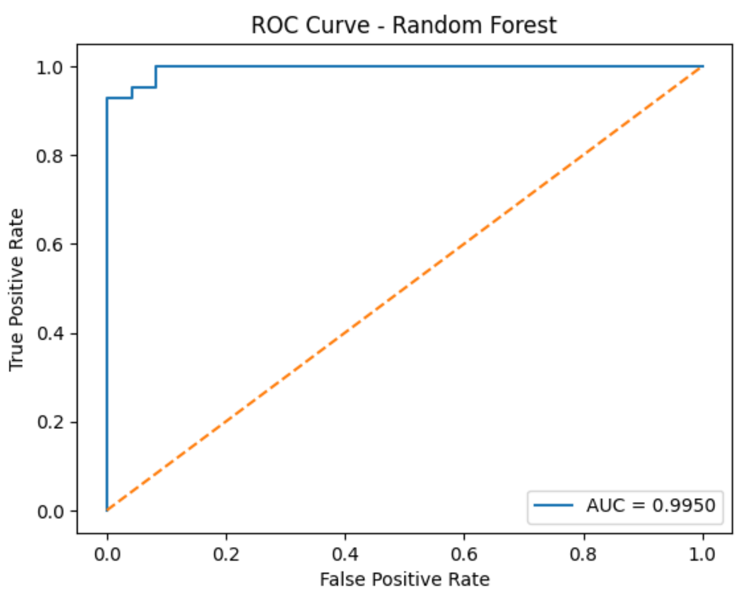

Figure 6: ROC Curve – Random Forest

Observation:

The Random Forest model produces fewer misclassifications compared to the Decision Tree. It achieves a higher AUC (approximately 0.97–0.99), demonstrating improved class separation and better generalization.

5.5 Combined ROC Curve Comparison

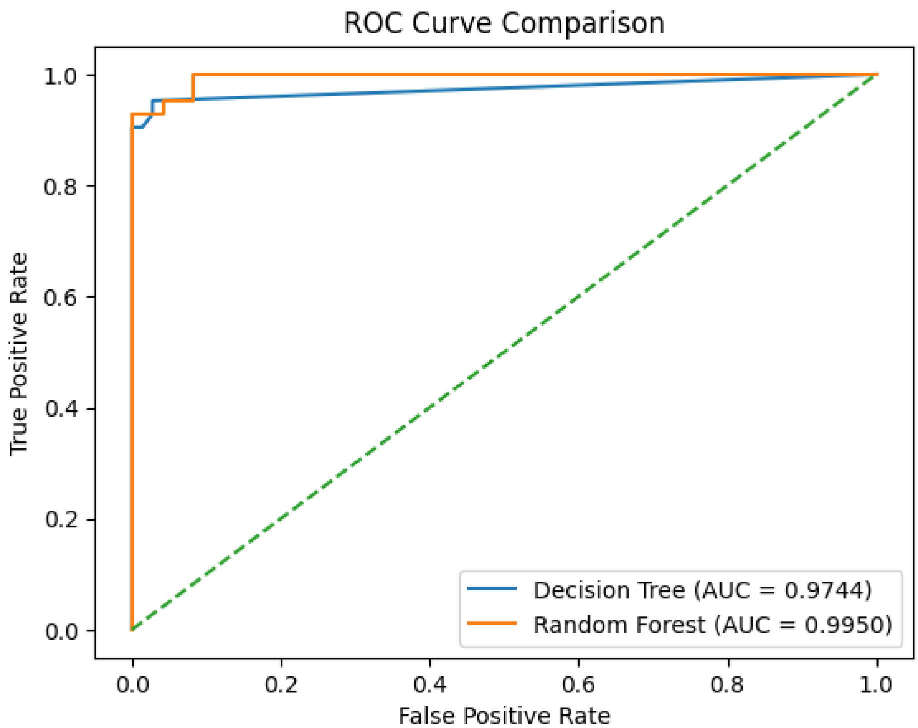

Figure 7: ROC Curve Comparison – Decision Tree vs Random Forest

Observation:

The ROC curve of the Random Forest consistently dominates that of the Decision Tree, indicating superior predictive performance and stronger generalization capability.

6. Hyperparameter Tuning Results

6.1 Table 1: Decision Tree Hyperparameter Evaluation (5-Fold CV)

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
Gini	3	92.08	0.88
Gini	5	90.76	0.87
Gini	7	92.30	0.89
Entropy	3	92.52	0.89
Entropy	5	92.30	0.89
Entropy	7	91.64	0.88

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

6.2 Table 2: Random Forest Hyperparameter Evaluation (5-Fold CV)

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
100	None	sqrt	96.48	0.952
100	None	log2	96.48	0.953
100	5	sqrt	96.70	0.955
100	5	log2	96.04	0.946
100	10	sqrt	96.48	0.952
100	10	log2	96.48	0.953
200	None	sqrt	96.26	0.949
200	None	log2	96.48	0.953
200	5	sqrt	95.60	0.940
200	5	log2	96.04	0.946
200	10	sqrt	96.04	0.946
200	10	log2	96.70	0.956

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

6.3 Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	92.98	86.84	88.59	93.85	92.92	91.04
Random Forest	96.49	93.85	96.49	96.49	96.46	95.95

Table 3: 5-Fold Cross-Validation Accuracy Comparison

7. Observation Questions

1. How does tree depth affect overfitting?

As the depth of the tree increases:

- Training accuracy generally increases.
- Validation accuracy improves initially but decreases after a certain depth.
- Very deep trees tend to memorize noise in the dataset, leading to overfitting.

A moderate tree depth provides the best balance between bias and variance, resulting in better generalization performance.

2. Which hyperparameter had the greatest impact?

For Decision Tree:

- `max_depth` had the most significant influence on performance.

For Random Forest:

- `n_estimators` significantly improved model stability.
- `max_features` reduced correlation among trees and enhanced generalization.

3. How does Random Forest improve generalization?

Random Forest improves generalization by:

- Using bootstrap sampling (bagging).
- Introducing feature randomness during tree construction.
- Averaging predictions from multiple decision trees.

These mechanisms reduce variance and prevent overfitting compared to a single Decision Tree.

4. Did ensemble learning always improve performance?

In this dataset, ensemble learning improved performance.
However, ensemble methods may not always improve results:

- If the dataset is very small.
- If base learners are underfitting.
- If noise dominates the signal.

8. Conclusion

Decision Tree and Random Forest classifiers were successfully implemented and evaluated using 5-Fold Cross-Validation.

Key Findings:

- Decision Trees are interpretable but prone to overfitting.
- Random Forest significantly reduces variance through ensemble learning.
- Cross-validation ensures robust hyperparameter selection.
- Random Forest achieved higher accuracy and AUC compared to Decision Tree.

Thus, ensemble learning improved model stability and generalization performance for breast cancer classification.